

Unit 4 . 1

- Abs/ local max/min
- Extreme value theorem
- Critical numbers
- Closed interval method
- Fermat's theorem (converse)

Unit 4 . 2

- Rolles' theorem
- Mean value theorem

Unit 4 . 3

- definition of an increasing/decreasing function
- Theorem: If $f' > 0$ then f increases and proof
- First derivative test
- What the derivative tells you from a sketch
- Concavity
- Second derivative test
- Inflection point